浙江大学 2024–2025 学年秋冬学期

《计算机组成与设计》课程期末考试试卷

课程号：67190020，开课学院：信息与电子工程学院

考试试卷：√A 卷、B 卷（请在选定项上打√）

考试形式：√闭、开卷（请在选定项上打√），

允许带 1 张 A4 纸大小的手写资料和计算器入场

考试日期：2024 年 1 月 2 日，考试时间：120 分钟

诚信考试，沉着应考，杜绝违纪。

考生姓名：_____学号：_____所属院系（专业）：_____

题序	一	二	三	四	五	总 分
得分						
评卷人						

I. CHOICE (Only one correct answer, 60 points)

1. The DRAMs are basically used as ( ) C

A: register

B: cache

C: main memory

D: disk

2. What is the decimal value of this 32-bit two's complement number? ( ) A

11111111 11111111 11111111 11111100

A: -4

B: -16

C: 4,294,967,292

D: -8

3. Assume x5 holds the value 0x0000000001010000. What is the value of x6 after the following instructions? ( ) B

```
        bge x5, x0, ELSE
        add x6, x5, x0
        jal x0, DONE
ELSE:   ori x6, x0, 2
DONE:
```

A: 80
B: 2
C: 160
D: 0

4. Consider the following code:

```
lb x6, 0(x7)
sb x6, 8(x7)
```

Assume that the register x7 contains the address 0×10000000 and the data at address is 0×1122334455667788.

What value is stored in 0×10000008 on a big-endian machine? ( ) A
A: 0x11
B: 0x22
C: 0x77
D: 0x88

5. What decimal number does the bit pattern 0xAC000000 represent if it is a two's complement integer? ( ) C
A: 201326592
B: 2885681152
C: -1409286144
D: -201326592

6. What is the minimum number of cycles needed to completely execute n instructions on a CPU with a k stage pipeline? ( ) B
A: n+k
B: n+k-1
C: n*k
D: n-k

7. Assume that x11 is initialized to 11 and x12 is initialized to 22. Suppose you executed the code below on a version of the pipeline introduced in our class that does **not** handle data hazards (i.e., the programmer is responsible for addressing data hazards by inserting NOP instructions where necessary). What would the final values of registers x14 be? ( ) A

```
addi x11, x12, 5
add x11, x11, x12
addi x14, x11, 15
```

A: 36
B: 42
C: 53
D: 33

8. How many NOP instructions should be added to the code below so that it will run correctly on a pipeline that does not handle data hazards? ( ) C

```
addi x11, x12, 5
add x13, x11, x12
addi x14, x11, 15
add x15, x13, x12
```

A: 1
B: 2
C: 3
D: 4

9. Dividing 101012 by 112, the quotient is ___and the reminder is___. C
A: 1002, 012
B: 1112, 012
C: 1112, 02
D: 1002, 02

10. A direct-mapped cache has 1K entries, cache line size is 2 words. What's the capacity of the cache? ( ) C
A: 2KiB
B: 4KiB
C: 8KiB
D: 16KiB

11. Assume that a virtual memory address is 32bit in length and the physical memory is 64MB. What's the size of a page table if page size is 32KB? ( ) D
A: 32KB
B: 64KB
C: 128KB
D: 512KB

12. Which of the following cache designer guideline is not valid? ( ) B
A: Fully associative caches have no conflict misses.
B: In reducing misses, associativity is more important than capacity.
C: The higher the memory bandwidth, the larger the cache block.
D: The shorter the memory latency, the smaller the cache block.

13. Assume the current PC is 0x10000010, what's next value of PC after execution of "jal x0, 100"? ( ) C
A: 0x00000064
B: 0x10000014
C: 0x10000074
D: 0x10000064

14. Assuming that there are 100 processors, what is the proportion of parallel part in parallel software to achieve an 80 speedup? ( ) A

A：99.75%

B：99.15%

C：98.75%

D：98.15%

15. For a 32-bit cache-memory system, a 32KB, 4-way set-associative cache has 2 words cache line size, how many bits are there in such cache's tag? ( ) A

A: 19

B: 21

C: 23

D: 25

16. For the following memory access pattern: 1,5,1,6,3, what's the content of a 4-entries, direct-mapped cache, assuming cache is vacant at the beginning. ( ) A

A:

	1	6	3

B:

1	6	3	5

C:

5	6	3	

D:

	6	3	

17. In a one level cache-memory system, assume that hit rate is 95%, and penalty to access the cache and main memory to be 5ns and 100ns respectively. We can infer that the average memory access time is ( ) ns. C

A: (5＋100)/2

B: 100×5%＋5×95%

C: 105×5%＋5×95%

D: 100×95%＋5×5%

18. Assuming forgetting to implement the hazard detection unit, how many cycles does it take to execute the following instruction sequence on a 5-stage pipeline with CPI(Cycle per Instruction) 1? ( ) C

```
I1:   lw   x1, 40(x6)
I2:   add x2, x3, x1
I3:   lw   x1, 16(x2)
I4:   sw   x2, 20(x4)
I5:   and x1, x1, x4
```

A: 8

B: 7

C: 6

D: 5

19. Consider a virtual memory system including only one-level page table. Suppose that it takes 150 ns to access the memory, 15 ns to access the TLB (i.e., Translation-lookaside buffer) and the hit rate of TLB is approximately 80 %. Thus, the average time to access the memory is ( ) ns. B
A: 185
B: 195
C: 205
D: 175

20. For a 32-bit cache-memory system, a 32 KB cache has 8 bytes cache line size. If the cache's tag needs 19 bits, the cache is ( ) C
A: direct-mapped
B: 2-way set-associative
C: 4-way set- associative
D: full-associative.

II. TRUE OR FALSE (10 points)

1.( ) A main memory access typically consumes less energy than a register file access.F

2. ( ) The memory hierarchy of a typical computer system comprises different memory technologies. T

3. ( ) A program with no branches has high temporal locality in its instruction memory references. F

4. ( ) A cache that has a block size equal to word size of memory access instructions cannot exploit spatial locality. T

5. ( ) The entire page table is typically stored in physical memory. F

6. ( ) An executable binary file that can run on a RISC-V CPU may not be able to execute on another RISC-V CPU. T

7. ( ) The CPI of superscalar processors can be more than one. T

8. ( ) Virtual-to-physical address translation is on the critical path of a memory access. T

9. ( ) Cache block size (B) can only affect miss latency. F

10. ( ) There is no way to reduce compulsory misses. F

III. Pipeline (10 points)

A certain workload having the following instruction mix is run on two processor designs with both having I-Cache and D-Cache.

ADD 10%
AND 20%
BEQ 25%
SD 15%
LD 30%

Additionally, it is known that I-Cache Hit-rate is 90%, D-Cache Hit-rate is 98%, 45% branches are not taken and 25% of LD instructions are followed by a dependent instruction. The memory takes 75 nano-seconds to access.

3.1 Assuming the above code is run on a standard 5-stage pipeline design processor with forwarding and hazard detection and with branches resolved in MEM and predicted not taken and clocked at 200MHz, what is the CPI? [4 points]

3.1225

Clock period : 1 / 200MHz = 5ns

Cache : 75 ns / 5ns = 15 cycles

CPI = 1 + 1*0.10*15 + (0.3+0.15)*0.02*15 + 0.3*0.25*1 + 0.25*0.55*3 = 3.1225

3.2 This five stage pipeline is extended to a similar 15 stage pipeline with no additional hazards being introduced. The amount of stall cycles needed for a ld followed by a dependent instruction does not change. The new frequency is 400MHz. Now the same code is run on the 15 stage pipeline where branches are resolved in the 11th stage.

a) What is the new CPI? [4 points]

b) Does this new design result in better performance for this workload? [2 points]

a)：

Beq : 11 – 1 = 10 cycles

Clock period : 1/400 MHz = 2.5ns

Cache : 75ns / 2.5 ns = 30 cycles

CPI = 1 + 1*0.1*30 + (0.3+0.15)*0.02*30 + 0.3*0.25*1 + 0.25*0.55*10 = 5.72

b)：

first : 5 ns * 3.1225 = 15.6125 ns ;

second : 2.5 ns * 5.72 = 14.3 ns ;

Yes, this new design result in better performance for this workload.

IV. CACHE (10 points)

You are trying to reverse-engineer the characteristics of a cache in a system, so that you can design a more e- cient, machine-specic implementation of an algorithm you are working on. To do so, you have come up with three sequences of memory accesses to various bytes in the system in an attempt to determine the following four cache characteristics:

- Cache block size (8, 16, 32, 64, or 128 B).
- Cache associativity (2-, 4-, or 8-way).
- Cache replacement policy (LRU or FIFO)

The only statistic that you can collect on this system is cache hit rate after performing each sequence of memory accesses. Here is what you observe:

Sequence	Addresses Accessed (Oldest → Youngest)	Hit Rate
1.	0→16→24→25→1024→255→1100→305	2/8
2.	31→65536→65537→131072→262144→8→305→1060	3/8
3.	262145→65536→4	2/3

Assume that the cache is initially empty at the beginning of the first sequence, but not at the beginning of the second and third sequence. The sequences are executed back-to-back, i.e., no other accesses take place in between sequences. Thus, **at the beginning of the second sequence, the contents are the same as at the end of the first sequence. At the beginning of the third sequence, the contents are the same as at the end of the second sequence.**

Based on what you observe, what are the following characteristics of the cache?

1) [4 points] Cache block size (8, 16, 32, 64, or 128 B)?
   ( )
2) [3 points] Cache associativity (2-, 4-, or 8-way)?
   ( )
3) [3 points] Cache replacement policy (LRU or FIFO)?
   ( )

Answer:

1)

16 B.

Explanation:
Cache hit rate is 2/8 in sequence 1. This means that there are 2 hits. Depending on the cache block size, we can group addresses that belong to the same cache block as follows:

- 8B: {0}, {16}, {24,25}, {255}, {305}, {1024}, {1100}. ∴ Number of possible hits = 1
- 16B: {0}, {16,24,25}, {255}, {305}, {1024}, {1100}. ∴ Number of possible hits = 2
- 32B: {0,16,24,25}, {255}, {305}, {1024}, {1100}. ∴ Number of possible hits = 3
- 64B: {0,16,24,25}, {255,305}, {1024}, {1100}. ∴ Number of possible hits = 4
- 128B: {0,16,24,25}, {255,305}, {1024,1100}. ∴ Number of possible hits = 5

Therefore, we can know that the cache block size is 16B.

2)
2-way
Explanation:
Cache hit rate is 3/8 in sequence 2, which means that there are 3 hits.
We already know that the cache block size is 16B. Thus, there are 4 offset bits.

The access to address 31 in sequence 2 would hit because the cache block would not be replaced.
The access to address 305 in sequence 2 would hit because the cache block would not be replaced.
The access to address 65537 in sequence 2 would hit because the cache block would not be replaced.

Therefore, all the other accesses should miss.
The access to address 65536, 131072 and 262144 in sequence 2 would miss because addresses 65536, 131072 and 262144 do not belong to any cache block previously accessed. Addresses 65536, 131072 and 262144 would be placed in set 0 if the cache associativity is 2-way, 4-way, or 8-way, independently of the cache size.
Address 8 must be a miss, so its cache block must be replaced by cache blocks that map to set 0 (addresses 65536, 131072 and 262144). For this to happen, the associativity must be 2-way.
Therefore, the cache is 2-way associative.

3)
FIFO.

From questions (a) and (b), we already know the following facts:

- The cache block size is 16 B.

- The cache is 2-way.

Cache hit rate is 2/3 in sequence 3, which means that there are 2 hits.
With the LRU policy only the access to address 262145 in sequence 3 would hit. With the FIFO policy, accesses to addresses 262145 and 4 in sequence 3 would hit.
Therefore, the cache adopts the FIFO policy.

V. VIRTUAL MEMORY (10 points)

5.1 Suppose we have a 48-bit address space with 32 GiB of physical memory and a 2 MiB page size. How many bits are in our page offset, physical page number, and virtual page number?[4 points]

Page offset: 21

Physical page number: 14

Virtual page number: 27

Regardless of your answers to Q5.1, now assume we have a 48-bit address space using 24-bit page offsets, 24-bit virtual numbers, and 12-bit physical page numbers. The system also has a TLB. The TLB and a subset of the page table is shown below. You may assume that the next physical page to be allocated has PPN 0x123, and that all accesses are independent of each other.

TLB			
Dirty	Valid	VPN	PPN
1	0	0x00 0000	0x203
0	1	0x00 0003	0x168
0	1	0x00 0002	0x164
1	0	0x61 C002	0x727
1	1	0x61 B001	0xC8E
0	0	0x0F 100F	0xE02
0	1	0x61 C001	0xAF4
1	1	0x00 0001	0x162

Page Table			
Index	Dirty	Valid	PPN
0x00 0000	1	1	0x161
0x00 0001	1	1	0x162
0x00 0002	0	1	0x164
…			
0x61 C000	0	0	0x625
0x61 C001	0	1	0xAF4
0x61 C002	1	0	0x727
…			

For each of the following virtual addresses, determine what will happen if we access this address. Choose the option you think is correct.

5.2 0x0000 00AB ACAB [2points]

( ) TLB hit
( ) TLB miss and page table hit √
( ) Page fault

5.3 0x61C0 02B1 ADE2 [2points]

( ) TLB hit
( ) TLB miss and page table hit
( ) Page fault √

5.4 0x61B0 01FE 3121 [2points]

( ) TLB hit √
( ) TLB miss and page table hit
( ) Page fault